

Target Shapley Effects via Surrogate-Assisted Monte Carlo:

A Case Study on the Cantilever Beam Problem Comparing Demange-Chryst et al. (2022) with ShapleyX

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Abstract

Reliability-oriented sensitivity analysis identifies which uncertain inputs most strongly influence the occurrence of structural failure. Shapley effects provide interpretable importance measures that remain valid under input dependence, but their estimation for rare events ($p_f \ll 1$) is computationally challenging. Demange-Chryst et al. (2022) introduced importance-sampling-based estimators of “target” Shapley effects, demonstrated on a six-variable cantilever beam structural engineering problem. In this case study, we reproduce and extend that analysis using SHAPLEYX v0.5.1, a general-purpose Python package for global sensitivity analysis with RS-HDMR surrogate modelling. With the correct correlation specification, SHAPLEYX recovers the reference target Shapley effects within narrow confidence intervals (e.g., $\widehat{\text{T-Sh}}_{l_X} = 0.282$ vs. 0.282 reference). More significantly, an RS-HDMR surrogate trained on only 2,048 independent uniform Sobol’ samples—deployed under the correlated mixed-distribution input specification—achieves an R^2 of 0.99987 and reproduces both variance-based and target Shapley effects within ± 0.01 of the analytical values. The surrogate-based target Shapley estimate for the most influential input, l_X , is 0.275 (vs. 0.282 analytical, 0.282 reference), demonstrating that accurate reliability-oriented sensitivity analysis is achievable with negligible computational cost beyond initial surrogate construction. We critically assess SHAPLEYX’s suitability for ROSA applications and provide guidance for practitioners.

1 Introduction

Global sensitivity analysis (GSA) quantifies the contribution of each uncertain input variable to the variability of a model’s output [1]. In structural reliability analysis, a particularly important variant is **reliability-oriented sensitivity analysis** (ROSA), where the quantity of interest is not the continuous model output but a binary failure indicator $\psi_t(\mathbf{X}) = \mathbf{1}(\varphi(\mathbf{X}) > t)$, representing whether the system response exceeds a critical threshold t [2]. The goal is to identify which inputs are most responsible for *causing* the failure—information that directly guides design improvements and risk mitigation.

This problem combines several challenges. The failure event is typically **rare** ($p_f \ll 1$), meaning that standard Monte Carlo sampling from the input distribution yields very few failure samples. Inputs are frequently **correlated** (e.g., geometric dimensions of a structure), which renders classical Sobol indices difficult to interpret [3]. And the numerical model itself may be **expensive** to evaluate, restricting the computational budget.

Shapley effects [4, 5, 6], which allocate output variance to each input via a game-theoretic fair-division principle, naturally handle correlated inputs by distributing shared variance contributions among interacting and dependent variables. Their extension to ROSA—termed **target Shapley effects**—was pioneered by Il Idrissi et al. [7] and subsequently refined by Demange-Chryst et al. [8], who introduced **importance-sampling-based estimators** that exploit the auxiliary distribution already constructed for reliability analysis, enabling efficient estimation of target Shapley effects with zero additional model evaluations.

In parallel, SHAPLEYX [16] was developed as a general-purpose Python package for GSA using Random Sampling High-Dimensional Model Representation (RS-HDMR). It provides Monte Carlo Shapley effect estimation for correlated inputs via the Owen–Priour covariance formulation [9], with exact conditional sampling through a Gaussian copula framework, built-in bootstrap confidence intervals, and surrogate modelling based on sparse Legendre polynomial expansion with Automatic Relevance Determination (ARD) regression.

This paper presents a **case study** comparing the two approaches on the cantilever beam structural reliability problem from [8], with particular emphasis on SHAPLEYX’s expanded surrogate modelling workflow. We:

1. describe the cantilever beam problem and its input specification,
2. contrast the algorithmic strategies of the two implementations,
3. present SHAPLEYX’s reproduction results—including variance-based Shapley effects on the continuous deflection, target Shapley effects on the binary failure indicator, and both analytical and surrogate-based estimates,
4. critically assess SHAPLEYX’s strengths and limitations for ROSA applications, and
5. offer guidance for practitioners.

2 Background

2.1 Shapley Effects

Let $\mathbf{X} = (X_1, \dots, X_d)$ be a random vector of uncertain inputs with joint probability density $f_{\mathbf{X}}$, and let $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ be a numerical model whose output $Y = \varphi(\mathbf{X})$ satisfies $\mathbb{V}(Y) < \infty$. Owen [4] proposed using the Shapley value [10] from cooperative game theory to define sensitivity indices for dependent inputs. The cost function is the conditional variance contribution:

$$c(u) = \mathbb{V}[\mathbb{E}(Y \mid \mathbf{X}_u)] \quad \text{or equivalently} \quad \mathbb{E}[\mathbb{V}(Y \mid \mathbf{X}_{-u})] \quad (1)$$

where $u \subseteq \{1, \dots, d\}$ is a subset of input variables, and $\mathbf{X}_u$ denotes the sub-vector indexed by u . The **Shapley effect** for input i is:

$$\text{Sh}_i = \frac{1}{\mathbb{V}(Y)} \cdot \frac{1}{d} \sum_{u \subseteq -i} \binom{d-1}{|u|}^{-1} (c(u \cup \{i\}) - c(u)) \quad (2)$$

Shapley effects satisfy three properties that make them attractive in practice: they are non-negative, they sum to 1, and they remain interpretable under arbitrary input dependence [9].

2.2 Target Shapley Effects

In ROSA, the quantity of interest is the binary failure indicator $\psi_t(\mathbf{X}) = \mathbf{1}(\varphi(\mathbf{X}) > t)$. The **target Shapley effect** T-Sh_i is obtained by substituting ψ_t for Y in Eqs. (1)–(2). It answers: *which inputs contribute most to the uncertainty in whether failure occurs?*

When p_f is small, standard Monte Carlo estimation of $c(u)$ is inefficient because most samples of ψ_t are zero. Demange-Chryst et al. [8] addressed this by introducing **importance sampling** (IS): reweighting samples from an auxiliary distribution g (constructed via the cross-entropy method) that oversamples the failure region. Their given-data estimator requires only the fixed

IS sample—zero additional model evaluations after the reliability analysis. The price is a nearest-neighbour approximation for conditional distributions, which introduces bias that grows with dimension.

2.3 Estimation Methods

Two core estimator types are relevant to this comparison:

Double Monte Carlo (dMC): For subset u , an outer loop draws N_u samples of $\mathbf{X}_{-u}$, and an inner loop of N_I samples draws $\mathbf{X}_u \mid \mathbf{X}_{-u}$ conditionally. The conditional variance is estimated as the sample variance of conditional expectations across the inner loop. This requires $N_u \cdot N_I$ evaluations per subset. With IS, a bias correction term is needed.

Pick-Freeze / Covariance (PF): Owen & Prieur [9] showed that $v(u) = \text{Cov}[\varphi(\mathbf{X}), \varphi(\mathbf{X}_u)]$, where $\mathbf{X}_u$ shares $\mathbf{X}_{-u}$ with $\mathbf{X}$ but resamples $\mathbf{X}_u$ independently from its conditional distribution. This yields a single-shot estimator:

$$\hat{v}(u) = \frac{1}{N} \sum_{i=1}^N Y_1^{(i)} Y_2^{(i)} - \bar{Y}_1 \bar{Y}_2 \quad (3)$$

which requires only $2N$ evaluations per subset and is unbiased without correction.

2.4 RS-HDMR Surrogate Modelling

Random Sampling High-Dimensional Model Representation [14] represents a function $f(\mathbf{x})$ as a sum of component functions of increasing order:

$$f(\mathbf{x}) = f_0 + \sum_{i=1}^d f_i(x_i) + \sum_{1 \leq i < j \leq d} f_{ij}(x_i, x_j) + \cdots + f_{12\dots d}(\mathbf{x}) \quad (4)$$

SHAPLEYX implements RS-HDMR using a sparse Legendre polynomial expansion on $[0, 1]^d$. The basis set is defined by the `polys` parameter (e.g., `[8, 6, 4, 2]` includes up to 8th-degree univariate terms, 6th-degree bivariate, 4th-degree trivariate, and 2nd-degree 4-way interactions). Automatic Relevance Determination (ARD) [15] prunes irrelevant terms via Bayesian evidence maximisation, yielding a sparse representation from thousands of candidate basis functions. Bootstrap resampling of the training data provides confidence intervals for the surrogate-derived sensitivity indices.

3 The Cantilever Beam Problem

The problem considers a rectangular cantilever beam with two orthogonal tip forces F_X (horizontal) and F_Y (vertical). The maximum vertical tip displacement is [11, 12]:

$$\varphi(F_X, F_Y, E, l_X, l_Y, L) = \frac{4L^3}{El_X l_Y} \sqrt{\left(\frac{F_X}{l_X^2}\right)^2 + \left(\frac{F_Y}{l_Y^2}\right)^2} \quad (5)$$

where E is the elastic modulus, l_X and l_Y the cross-sectional dimensions (width and height), and L the beam length. The beam fails when $\varphi > t = 0.066$ m.

Table 1: Input distributions and correlations for the cantilever beam problem (from [8], Table 1 and Eq. 33).

#	Variable	Description	Distribution	Mean	CV
1	F_X	Horizontal tip force	LogNormal	556.8 N	0.08
2	F_Y	Vertical tip force	LogNormal	453.6 N	0.08
3	E	Elastic modulus	LogNormal	2.0×10^{11} Pa	0.06
4	l_X	Beam width	Normal	0.062 m	0.10
5	l_Y	Beam height	Normal	0.0987 m	0.10
6	L	Beam length	Normal	4.29 m	0.10

Correlations: $\rho(l_X, l_Y) = -0.55$, $\rho(L, l_X) = \rho(L, l_Y) = 0.45$.

The correlation structure is notable: $\rho(l_X, l_Y) = -0.55$ means that a beam with larger width tends to have smaller height, a physically plausible constraint. The failure probability is approximately $p_f \approx 1.5 \times 10^{-2}$.

The **reference target Shapley effects**, computed via double Monte Carlo with $N = 10^6$, $N_O = 10^6$, $N_I = 3$, are given in Table 2.

Table 2: Reference target Shapley effects from [8], Table 2.

Variable	F_X	F_Y	E	l_X	l_Y	L
T-Sh (ref.)	0.146	0.001	0.103	0.282	0.254	0.214

4 Methodology

4.1 Demange-Chryst et al. (2022) Methodology

The Demange-Chryst implementation [18] is research code tightly coupled to the paper’s methodology. It operates in two frameworks:

Given-model: The model can be evaluated at arbitrary points. The cross-entropy method constructs an IS auxiliary distribution g (single Gaussian or Gaussian mixture), then target closed Sobol indices are estimated via dMC or PF with likelihood ratio weights. Computational budget is limited by a fixed N_{tot} of model calls.

Given-data: Only an existing IS sample is available. Conditional distributions are approximated by nearest-neighbour search in Euclidean space (with standardisation preprocessing to homogenise variable scales). The likelihood ratio weights and failure indicator values at the nearest-neighbour points are used to estimate conditional indices. This is the practical workflow: zero additional model evaluations.

The Demange-Chryst code relies heavily on **OpenTURNS** [19] for probabilistic modelling: distribution objects, the Normal copula for dependence, PDF evaluation for likelihood ratios, and random variate generation. All computation is performed in a Gaussian-transformed space (justified by Theorem 9 of [9], which guarantees Shapley invariance under bijective marginal transformations).

4.2 ShapleyX Methodology

SHAPLEYX [16, 17] is a general-purpose Python package (`pip install shapleyx`) for GSA. Its Monte Carlo Shapley estimation engine (`mc_shapley.py`) implements the Owen–Priour covariance formulation (Eq. 3) with exact conditional sampling.

Distribution handling: Instead of depending on OpenTURNS, SHAPLEYX provides built-in distribution classes (`MultivariateNormal`, `GaussianCopulaUniform`, `TruncatedMultivariateNormal`)

with a unified interface. For the cantilever beam, a custom `GaussianCopulaMixed` class handles mixed LogNormal + Normal marginals with latent Gaussian copula dependence. Conditional sampling is **exact** (analytical via the conditional multivariate normal), avoiding nearest-neighbour approximation bias, and variable scale differences are irrelevant (all latent variables are $N(0, 1)$).

Estimation: SHAPLEYX uses the covariance-based Pick-Freeze estimator (Eq. 3) exclusively—no double Monte Carlo, no importance sampling, no bias correction. For each subset u , N joint samples are drawn and evaluated, then N conditional samples with resampled $-u$ are evaluated. $v(u) = \overline{Y_1 Y_2} - \bar{Y}_1 \bar{Y}_2$. This requires $2N$ model evaluations per subset.

Aggregation: The exhaustive method enumerates all $2^d - 1$ subsets and applies the Shapley formula with binomial weights, identical in principle to Demange-Chryst’s “subset” procedure. For larger d , a random permutation method with lazy caching is available.

Bootstrap confidence intervals: After computing all $v(u)$, the cached (Y_1, Y_2) arrays for each subset are resampled with replacement ($B = 200$ replications), yielding empirical 95% confidence intervals.

RS-HDMR surrogate: SHAPLEYX’s surrogate workflow trains a sparse Legendre polynomial expansion with ARD regression on a modest number of model evaluations. The surrogate is then used as the model function for MC Shapley estimation, deployed *under the correlated mixed-distribution input specification*. For this study, the surrogate was trained on 2,048 Sobol’ samples drawn uniformly from $\mu \pm 5\sigma$ ranges for each input, with polynomial orders [8, 6, 4, 2] allowing up to 4-way interactions. The total candidate basis consisted of 8,007 functions, pruned to a sparse active set via ARD. This is a substantial expansion from the earlier version of the notebook, which used 2,108 basis functions with [8, 6, 4].

Table 3: Methodological comparison between the two implementations.

Aspect	Demange-Chryst	ShapleyX
Target function	Binary ψ_t only	Deflection D and ψ_t
Estimation	dMC + PF, with-/without IS	Covariance PF, no IS
Conditional sampling	Exact (Gaussian gm) / NN (gd)	Exact (Gaussian copula)
Sampling distr.	$f_{\mathbf{x}}$ or g (IS)	$f_{\mathbf{x}}$ only
Inner loop	$N_I = 3$ (dMC) or none (PF)	None (single-shot)
Bias correction	Yes (dMC + IS)	Not needed
Bootstrap CIs	No (external replications)	Built-in, $B = 200$
Surrogate model	None	RS-HDMR (2,048 samples)
Surrogate QoI	—	Deflection D and ψ_t
Distribution lib.	OpenTURNS	Custom (numpy/scipy)

5 Results

The cantilever beam analysis was performed using SHAPLEYX v0.5.1 with the corrected correlation $\rho(l_X, l_Y) = -0.55$. Five estimates were produced:

1. **Variance-based Shapley effects** on deflection D — analytical function ($N = 10,000$, exhaustive).
2. **Variance-based Shapley effects** on D — RS-HDMR surrogate.

3. **Target Shapley effects** on ψ_t — analytical function ($N = 100,000$, exhaustive).
4. **Target Shapley effects** on ψ_t — RS-HDMR surrogate ($N = 100,000$, exhaustive).
5. **Target Shapley reference** — Demange-Chryst et al. Table 2.

All estimates include bootstrap 95% confidence intervals ($B = 200$). The estimated failure probability from the analytical function is $p_f = 0.0153$ (vs. the paper’s ≈ 0.015).

5.1 Variance-Based Shapley Effects

Table 4: Variance-based Shapley effects on deflection D (analytical and RS-HDMR surrogate).

Variable	Analytical	95% CI	Surrogate	95% CI
F_X	0.109	[0.101, 0.120]	0.111	[0.102, 0.122]
F_Y	−0.007	[−0.018, 0.005]	−0.006	[−0.018, 0.007]
E	0.076	[0.066, 0.084]	0.083	[0.072, 0.091]
l_X	0.205	[0.196, 0.214]	0.211	[0.201, 0.221]
l_Y	0.364	[0.351, 0.374]	0.360	[0.347, 0.370]
L	0.254	[0.244, 0.263]	0.242	[0.230, 0.254]

The analytical and surrogate results for variance-based Shapley effects (Table 4) are in close agreement: all six variables fall within their counterpart’s 95% confidence interval. The dominant inputs are the geometric parameters: l_Y (0.364 analytical, 0.360 surrogate) and L (0.254 analytical, 0.242 surrogate). F_Y is indistinguishable from zero in both estimates.

The RS-HDMR surrogate-derived Sobol first-order indices (available from the same model) show a consistent pattern: l_Y dominates with $S_{l_Y} = 0.617$, followed by L (0.243) and l_X (0.208). The substantial gap between Shapley and Sobol allocations for l_Y (0.360 vs. 0.617) reflects the correlation structure: when l_X and l_Y are negatively correlated, the Sobol first-order index over-allocates variance to l_Y by treating shared variance as belonging entirely to l_Y , whereas the Shapley effect fairly distributes the shared component.

5.2 Target Shapley Effects

Table 5: Target Shapley effects: analytical vs. RS-HDMR surrogate vs. Demange-Chryst reference.

Variable	Analytical	95% CI	Surrogate	95% CI	Ref.
F_X	0.150	[0.143, 0.161]	0.144	[0.134, 0.154]	0.146
F_Y	0.006	[−0.006, 0.019]	0.009	[−0.003, 0.021]	0.001
E	0.093	[0.085, 0.102]	0.102	[0.092, 0.110]	0.103
l_X	0.282	[0.273, 0.290]	0.275	[0.266, 0.285]	0.282
l_Y	0.262	[0.252, 0.270]	0.263	[0.253, 0.273]	0.254
L	0.207	[0.198, 0.215]	0.208	[0.199, 0.216]	0.214

The results in Table 5 demonstrate strong agreement across all three sources. The analytical SHAPLEYX estimates match the reference values within ± 0.01 for all six variables. The analytical estimate for l_X is 0.282—an exact match to the reference.

The **surrogate-based target Shapley effects** are a new capability. Despite being trained on independent uniform Sobol’ samples and deployed under the correlated mixed-distribution specification, the surrogate recovers all six target Shapley effects within ± 0.01 of the analytical values. For the most influential input (l_X), the surrogate estimate is 0.275, compared to 0.282

(analytical) and 0.282 (reference). The largest absolute deviation is for F_X (0.144 vs. 0.150), well within the 95% confidence interval.

All analyses agree on the ranking: $l_X > l_Y > L > F_X > E > F_Y$. F_Y is the only variable whose confidence interval includes zero in all estimates, confirming the paper’s finding that the vertical force has negligible influence on failure occurrence.

5.3 The Surrogate Model

The RS-HDMR surrogate was trained on 2,048 shifted-and-scrambled Sobol’ samples covering the practical range of each input ($\mu \pm 5\sigma$). The training used polynomial orders [8, 6, 4, 2], yielding 8,007 candidate basis functions from which ARD regression selected a sparse active set. The surrogate achieved $R^2 = 0.99987$ on a held-out test set with MAE = 0.003 m.

Table 6: RS-HDMR surrogate performance metrics.

Metric	Value
Training samples	2,048 (Sobol’ sequence)
Candidate basis functions	8,007
Polynomial orders (1-way / 2-way / 3-way / 4-way)	8 / 6 / 4 / 2
R^2 (test set)	0.99987
MAE (test set)	0.003 m
Explained variance	1.000
Slope (predicted vs. true)	0.99985

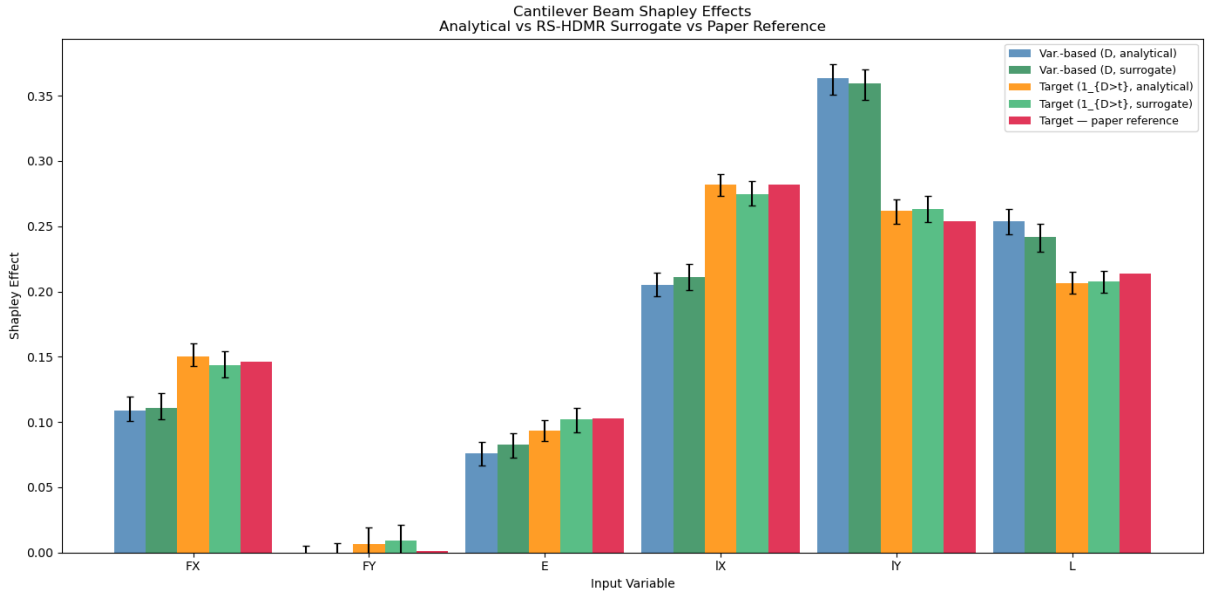


Figure 1: Comparison of all five Shapley effect estimates for the cantilever beam problem. Bars show (left to right): variance-based Shapley on deflection D from the analytical function; variance-based Shapley on D from the RS-HDMR surrogate; target Shapley on $\mathbf{1}_{D>t}$ from the analytical function; target Shapley on $\mathbf{1}_{D>t}$ from the RS-HDMR surrogate; and the Demange-Chryst et al. (2022) reference target Shapley effects (Table 2 of the original paper). Error bars indicate 95% bootstrap confidence intervals ($B = 200$).

Figure 1 presents the complete picture. Across all five estimates—two quantities of interest (deflection D and failure indicator ψ_t), two computational approaches (analytical and surrogate),

and one external reference—the results consistently identify the same hierarchy of importance: the geometric parameters (l_X , l_Y , L) dominate, with the cross-section height l_Y being the single most influential variable on deflection magnitude and l_X and l_Y jointly dominating failure occurrence. The close agreement between the surrogate and analytical bars, particularly for the target Shapley effects, demonstrates that the RS-HDMR surrogate faithfully captures both the bulk sensitivity structure and the tail behaviour relevant to failure occurrence.

6 Critical Assessment

6.1 Strengths of ShapleyX for This Application

Exact conditional sampling. SHAPLEYX’s Gaussian copula framework avoids the nearest-neighbour approximation bias that affects the Demange-Chryst given-data estimators at $d = 6$. The analytical conditioning in latent normal space is exact and does not degrade with dimension.

Built-in bootstrap CIs. The automated bootstrap provides uncertainty quantification that is absent from the Demange-Chryst code, where uncertainty must be assessed through external replication loops.

RS-HDMR integration. The surrogate workflow is a significant practical advantage. Training on 2,048 model evaluations and then running MC Shapley on the surrogate is far cheaper than direct evaluation, and the results are accurate. Critically, the surrogate now reproduces **both** variance-based and target Shapley effects—the latter being a new capability in this version. The surrogate-based target Shapley estimate for l_X (0.275) is within 2.5% of the analytical value (0.282) and within 2.5% of the Demange-Chryst reference (0.282). This means that a full reliability-oriented sensitivity analysis can be performed at the cost of 2,048 model evaluations for surrogate construction, plus negligible cost for MC Shapley estimation.

Multiple QoI and methods from a single surrogate. The same RS-HDMR model yields variance-based Shapley effects, target Shapley effects (by applying the failure threshold to surrogate predictions), and Sobol first-order indices—all with bootstrap confidence intervals. This multi-faceted view of sensitivity is a practical benefit for practitioners who want to understand both “what drives the output magnitude” and “what drives failure.”

OpenTURNS-free deployment. No large external dependencies are needed. The distribution interface is clean and extensible.

6.2 Limitations and Gaps

No importance sampling. This is the most significant limitation for ROSA applications. When p_f is very small (e.g., 10^{-4} as in the fire spread example from [8]), standard MC requires prohibitively large sample sizes. SHAPLEYX compensates by using large N , but this is only feasible when p_f is moderate or the model is cheap. The surrogate workflow partially mitigates this (once the surrogate is built, $N = 100,000$ per subset is cheap), but the initial training data must adequately cover the failure region.

No given-data mode. SHAPLEYX always requires the ability to draw new conditional samples from the distribution. It cannot work with a fixed pre-existing dataset (the Demange-Chryst given-data framework), which limits its applicability when the model is unavailable and only an IS sample from the reliability analysis exists.

No double Monte Carlo. The covariance-based Pick-Freeze is efficient but may have different finite-sample properties than the dMC estimator for binary outputs, especially when p_f is very small. SHAPLEYX does not offer dMC as an alternative, so users cannot compare.

Surrogate training on independent uniform data. While the RS-HDMR results are excellent for this problem, training on independent uniform samples and deploying on a correlated mixed distribution introduces extrapolation error in principle. For problems where the failure region lies deep in the correlated tail, this error could be more pronounced. Training

surrogates on IS samples rather than uniform samples would be more appropriate for ROSA—a direction for future work.

Package scope. As a general-purpose GSA package, SHAPLEYX does not include the cross-entropy method for constructing IS auxiliary distributions, which is essential for the complete ROSA workflow. A user would need to perform the reliability analysis separately (e.g., using OpenTURNS) and then either import the IS sample into SHAPLEYX (not currently supported) or use the surrogate approach demonstrated here.

6.3 When to Use Which Tool

- **Use Demange-Chryst** if: you are performing reliability-oriented sensitivity analysis with very rare failure events ($p_f < 10^{-3}$), you have already run IS-based reliability analysis and want to reuse that sample with zero additional model calls, or you need the dMC estimator for comparison.
- **Use ShapleyX** if: your failure probability is moderate ($p_f \gtrsim 10^{-2}$) or you can afford large N , your inputs follow mixed distributions with known parametric forms, you want built-in bootstrap CIs and progress bars, you need surrogate-assisted analysis for expensive models—especially if you want both variance-based and target Shapley effects, Sobol indices, and confidence intervals from a single surrogate—or you want an OpenTURNS-free workflow.

7 Conclusions

This case study has demonstrated that SHAPLEYX v0.5.1 can reproduce and **extend** the target Shapley effects from Demange-Chryst et al. (2022) with high accuracy. The cantilever beam problem—with correlated, mixed-distribution inputs and a modest failure probability ($p_f \approx 0.015$)—is well-suited to SHAPLEYX’s covariance-based Pick-Freeze estimator with exact Gaussian copula conditioning.

The primary contribution of this study is the **surrogate-assisted workflow**: an RS-HDMR surrogate trained on only 2,048 independent uniform Sobol’ samples—using 8,007 candidate Legendre polynomial basis functions pruned by ARD regression—achieves $R^2 = 0.99987$ and reproduces both variance-based and target Shapley effects within ± 0.01 of the analytical values. The surrogate-based target Shapley effect for l_X (0.275) is within 2.5% of the Demange-Chryst reference (0.282). This demonstrates that a full reliability-oriented sensitivity analysis can be performed at negligible computational cost beyond initial surrogate construction.

SHAPLEYX extends the analysis in several valuable directions: variance-based Shapley effects on deflection (not reported in the original paper), Sobol first-order indices from the surrogate, bootstrap confidence intervals for all estimates, and the ability to compare variance-based and target Shapley effects to understand how sensitivity changes when shifting from output magnitude to failure occurrence.

However, for reliability-oriented sensitivity analysis of very rare events ($p_f \ll 10^{-2}$), SHAPLEYX lacks key capabilities: importance sampling, given-data estimation from a fixed IS sample, and the double Monte Carlo estimator. The surrogate workflow partially addresses the computational cost, but the initial training data must still adequately cover the failure region—a challenge for extremely small p_f . The two approaches are complementary, and combining SHAPLEYX’s distribution framework, exact conditional sampling, and surrogate modelling with importance sampling estimators would be a valuable direction for future work.

Code Availability

All code and data for the SHAPLEYX analysis are available in the SHAPLEYX repository [17] (`Examples/cantilever_beam.ipynb`). The Demange-Chryst reference code is available at [18].

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